

MAJOR PROJECT REPORT

On

Food Nutrition Estimator from Image and Text using AI

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Under the guidance of:

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DECLARATION

I, Aryan Walia student of B.Tech CSE (IoT and Cybersecurity including BCT) hereby declare that the Major Project Report titled "LiveFit — AI-Powered Nutrition & Meal Management Mobile Application", submitted to the Department of Computer Science and Engineering, Amity School of Engineering and Technology, Amity University Uttar Pradesh, Noida, in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering, has not previously formed the basis for the award of any degree, diploma, or other similar title or recognition.

Place: NOIDA

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CERTIFICATE

On the basis of the declaration submitted by ARYAN WALIA, student of B.TECH Computer Science and Engineering (Internet of Things and Cybersecurity including Blockchain Technology), I hereby certify that the Major Project titled "LiveFit — AI-Powered Nutrition & Meal Management Mobile Application" submitted to the Department of Computer Science and Engineering, Amity School of Engineering and Technology, Amity University, Noida, in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is an original contribution with existing knowledge and faithful record of work carried out by them under my guidance and supervision.

To the best of my knowledge, this work has not been submitted in any part or full for any degree or diploma to this university or elsewhere.

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Abstract

LiveFit is an all-inclusive, full-stack, nutrition and meal management solution based on AI-enhanced technology that aims at closing the gap between food awareness and individual dietary recommendations. In a time where the non-communicable diseases that are associated with a bad diet are growing drastically particularly in urban India, users are increasingly demanding smart, convenient, and personalised applications to control their eating habits. LiveFit also fills this requirement by integrating a cross platform mobile application (written in React Native and Expo), a modern web app, and a powerful cloud-hosted platform into a unified ecosystem. The platform itself simplifies and is based on an intelligent food identification engine that allows one to enter the name of the food or take a photo and then it identifies the food. On submission of an image a hybrid machine-learning pipeline is called, in which a locally-hosted Python FastAPI microservice serves the Hugging Face transformer model 'rajistics/finetuned-indian-food' which is specially trained to recognise Indian dishes. When the model is below 60 per cent in confidence, the system transparently reverts to the Roboflow computer vision API which is able to identify the unusual and ambiguous food photographs strongly. The name of the identified food is then matched against a prioritised, dual, database, nutritional look up strategy. In the case of Indian food, the system interrogates an Indian Food Composition Tables (IFCT) dataset of exact, partial, and reverse fuzzy matched database giving very precise macronutrient data of thousands of subcontinental foods. On the rest of the food products, the system makes a query to the USDA FoodData Central API which is an international authoritative nutritional database.

In addition to food logging, LiveFit also has Google Gemini 2.5 Flash, which is a conversational nutrition coach based on AI. This AI assistant receives the real-time data regarding the nutritional intake of the user and provides a personalised and acting dietary advice to the natural language questions, recognises macro deficiencies or excesses and provides the appropriate food options. The platform also includes a reward ecosystem that is gamified to incentivize regular food logging whereby the user earns LiveFit Coins, which can be then used to purchase digital items such as discount codes based on percentage, free meals, Pro subscriptions, and nutrition reports in a dedicated store. The discount codes are directly coded with Razorpay payment gateway and gives the user the option to use authenticated coupon codes at checkout where the price is updated real-time. There is a level of subscription (Free and Pro) that opens special capabilities such as an unlimited AI coach interaction and better analytics. The meal ordering part will enable them to view a recommended list of health-optimised meals, add to a permanent cart, and reserve the purchase with the help of Razorpay, and PDF invoices will be automatically created at the time of order placement. All user information is safely stored within a cloud MongoDB Atlas database, its authentication is done with the help of bcrypt password hashing and the stateless session management on the application is enforced under the use of the JSON Web Tokens.

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Chapter 1: Introduction

1.1 Background

Scientific knowledge of the connection between diet and health has never been as clear, but to apply nutritional knowledge in everyday life is one of the most challenging. perennial social health issues of the modern world. The World Health has indicated that it is. An estimated 2.8 million people are killed every year as a result of being. Obese or overweight, and there is an additional colossal load which is ascribed to micronutrient. nutritional deficiencies and diet related non-communicable illnesses such as diabetes, cardiovascular disease, and some cancers [27]. The dual burden in India particularly. obesity and metabolic syndrome with their rapid growth expose a singular. complicated dietary environment and requires culturally competent and technologically advanced solutions. Conventional food diary techniques such as paper food diaries, as well as generic calorie. tallying tables, and universal nutritional guidelines have been shown to be systematic. inadequate. They are cumbersome, inaccurate, culturally ill-illustrated to the extraordinary. diversity of Indian food, and is essentially out of touch with the live feedback. circles that have been provided by modern behavioural science to be fundamental in sustainable habit. formation. The rapid growth in using smartphones particularly among the rapidly increasing population in India. growing middle, offers a historic challenge to implementation of intelligent, individualised nutritional applications at the population level.

LiveFit was the idea created to bridge this gap [4][5]. Instead of it presenting a fixed. LiveFit is an active nutrition-based smart weighing system that fulfills the needs of calorie counting. wherever they are, users can capture a photo of a thali they cooked at home or find the location of the thali around them. contain macronutrients in a packaged food product, and is responsive to an immediate reaction. individualised, and contextualised nutritional advice. The site is created in a user-friendly way. better be an Indian-first ground-up to allow the IFCT nutritional database and a fine. used in tuned food recognition of Indians, but perfectly able to operate on global. the cuisines via the USDA integration.

1.2 Problem Statement

Although the work on health and fitness applications is gaining momentum in recent years, a substantial portion of the population, especially Indian people, remains to encounter significant obstacles on their way to the correct monitoring of nutritional intake. The obstacles have a multi-dimensional nature. To start with, there is the issue of coverage of Indian food. Even the nutrition applications currently in place are constructed on Western food databases, especially the USDA database, which hardly covers thousands of individual Indian dishes eaten daily all over the subcontinent. One trying to enter dal makhani, 'aloo paratha', or 'rasam' will frequently find either no match, an incorrect match, or nutritional data that bears little relationship to the actual composition of the dish as typically prepared in India. Second, one is the manual food logging. It has been repeatedly shown by research that the cognitive and temporal burden of the very process of dietary tracking is the sole most predictive factor of dietary tracking adherence [4][6]. How many of them will abandon

the use of tracking applications in days of installation after having to manually search and select the foods, manually estimate the portion sizes, and key-type in the quantities?

Third, there is the lack of smart and personalised feedback. The vast majority of nutrition apps are little more than a ledger book, holding what has been eaten, without applying actually useful advice on what a change would be best helping the health interests of the application user. In the absence of this intelligent feedback loop the motivational value of tracking data decays very quickly.

Fourth, there is the challenge of sustaining user engagement over time. Health behaviour change is inherently a long-term project, and applications that fail to maintain user engagement beyond the initial novelty phase cannot deliver meaningful health outcomes. Gamification, social features, and reward mechanisms have been shown in the literature to significantly improve long-term engagement, but few nutrition applications implement these mechanisms in a sophisticated or genuinely rewarding way.

LiveFit addresses all four of these problems through an integrated system architecture that prioritises Indian food recognition, minimises logging friction through AI-powered identification, provides real-time intelligent coaching through generative AI, and sustains engagement through a meaningful coin and reward ecosystem.

1.2.1 The Indian Nutrition Gap in Detail

India's nutritional diversity is staggering. With over 7,500 distinct dishes documented across 28 states and 8 union territories, each with unique ingredients, preparation methods, and regional variants, no single Western-designed database can adequately serve an Indian user. Consider the simple case of 'sambar': the Tamil Nadu version differs markedly in composition from the Karnataka or Kerala variant, and the homecooked version differs further still from restaurant preparations. Caloric estimates derived from generic databases may err by 30–50% for such dishes, systematically corrupting the dietary records of millions of users who rely on these applications for health management.

This issue is further exacerbated by the organization of typical Indian cuisine, which hardly fits into the discrete groupings to which database-based applications are more accustomed. An average North Indian thali has six to ten items at once, of which a large number are cooking specific combinations of items having no database representation. Similar problems are faced by South Indian tiffin, Bengali fish curries and Gujarati farsan. Such a structural dislocation between the Indian dietary pattern and the presuppositions used in the current digital tracking tools is the largest obstacle to the use of nutrition applications amongst the Indian marketplace.

1.2.2 The Engagement Deficit in Health Applications

In addition to the question of accuracy, there is a long-reported engagement crisis in the health application industry. The industry records have been consistent in revealing that more than 8 out of ten applications users of the health and fitness apps drop their application of choice within the first month of installation. The reasons are multifactorial, such as the

friction of manual data entry, the lack of meaningful progress feedback, the lack of social accountability mechanisms, and the inability to give tangible rewards to engage in the sustained engagement. Successful applications, which continue to attract the interest of customers over time, always have these elements in common: sleek interface designs, smart personalisation, purposeful progressions, and concrete reward models. LiveFit is built with an initial aim of having all these four retention mechanisms.

The gap in economic aspects of this involvement is also notable. Users who lack the motivation to continue with nutrition applications are usually those who do not find the perceived cost of using the applications to offer more than the perceived value. It is especially acute during the first two to four weeks of the utilization before any significant health outcome could be observed. The only solution to this disparity is to create instant, tangible value every time you interact with someone an idea reinforced by the practice of rewarding LiveFit members with a coin whenever they log food, so that each tracking behaviour would create an instant, visible, and economically significant benefit.

1.3 Proposed Solution

Livefit suggests a complete hybrid solution that will combine computer vision, natural language AI, gamified reward and curated database of nutritional data into one user experience. The solution has been designed to work on two supplementary surfaces: a mobile app implemented in React Native and Expo to log meals on the go and coach and a web app implemented in vanilla HTML, CSS, and JavaScript to have more sophisticated experiences of browsing and management.

The technical centre of the system is the food identification line. Upon uploading a food photograph, this is sent to a Python FastService that is locally hosted and serves a finetuned vision transformer model directly trained on the categories of Indian food. It is supported by instant and low-latency reactions with zero per-ask API expenses on the most frequent query. By using a confidence threshold mechanism, the local model only accepts the high confidence prediction, otherwise the local model activates an automatic fallback to an external fallback Roboflow computer vision API that offers a strong safety net to weird foods or photographs of less-than-optimal quality.

The nutritional resolution layer puts into effect a prioritised dual-database strategy. The embedded IFCT data is first searched using the identified Indian food items and includes detailed macronutrient profiles of traditional Indian foods. The non-Indian (or Indian food not available in IFCT) foods are matched against the USDA Food data Central API. This is a stratified method that would maximise precision of the Indian cuisine, and generous coverage of international foods.

Table 1.3.1: Core Components of LiveFit

Feature	Implementation	User Benefit
Food ID by Image	HuggingFace + Roboflow hybrid ML pipeline	Zero-friction photo-based logging
Food ID by Text	IFCT + USDA dual-database lookup with fuzzy matching	High accuracy for Indian & global foods
AI Coaching	Google Gemini 2.5 Flash with real-time nutrition context	Personalised dietary advice 24/7
Gamification	Coin earning + redemption reward store	Sustained long-term engagement
Meal Ordering	Curated health-focused menu + Razorpay payment	Healthy food delivery integration
Subscriptions	Free / Pro tiers via Razorpay	Monetisation with premium features
Secure Auth	bcrypt + JWT + email reset	Enterprise-grade account security

1.4 Objectives

- The project intends to meet the following main objectives that seek to fill a certain gap in the existing sphere of nutrition technologies applications
- Build a complete functionality nutritional tracking system with mobile and web interfaces, so that any type of user can use the entire set of functionalities.
- Installed a potentially effective pipeline. A high-accuracy hybrid food recognition pipeline between local fine-tuned machine learning and cloudcomputer-vision fallback providing high reliability food recognition of local Indian cuisines and globally.
- Implement nutritional resolution system with a dual-database platform (using the IFCT dataset of Indian food and the USDA Fooddata central API of global food) to give precise data on macronutrients of thousands of food products.

- Implement a text-based AI nutrition coaching application based on Google Gemini 2.5 Flash, which can give the user customised nutrition information depending on the real-time data of his/her nutrition.
- Develop and deploy a gamified coin and reward ecosystem that offers rewards to long-term food logging and ordering healthy meals to increase user retention and loyalty to the platform.
- Combine fully functional payment system with options to purchase meals and upgrade subscription with Razorpay, and automatically create PDF receipts on all accepted orders.
- Make sure that the entire platform is secure with the use of the password hashing algorithm BCrypt, stateless authentication using JWT and the communication between the APIs being secure.

1.4.1 Technical Objectives

Technically, the project will aim to show that, with entirely open-source software and free-tier cloud service usage, it is possible to contemplate the construction of a production-quality AI-powered health platform and deploy it, which would mean that the approach to architecture can subsequently be replicated and used in the future by other researchers and commercial developers. The system in particular addresses the following technical goals: reaction to queries based on the images in less than 2 seconds, accuracy in nutritional lookup of over 90 percent in response to common Indian dishes, accuracy in AI coaching response at the level of rule-based dietary advice systems, and payment collection rate that is on par with the rest of the Razorpay platform.

1.4.2 Research Objectives

In the framework of the research perspective, the project will help to confirm various empirical hypotheses, which have far-reaching implications to the overall sphere of digital health. To begin with, the fact that fine-tuned domain-specific vision transformer is superior to general-purpose computer vision APIs in recognizing Indian food. Second, the relevance and accuracy of dietary advice is significantly improved by such real-time contextual grounding of responses of large language models, which is quantifiable. Third, the gamification functions based on physical economic incentives enhance food logging adherence in short-term much better than applications based on passive tracking. They are tested with the help of the functional prototype and assessed using structured test cases in Chapter 6.

1.4.3 Social and Health Objectives

On the biggest scope, LiveFit would help increase the nutritional-related health results of the urban Indian population by ensuring that proper and personalised dietary recommendations are available to any person who has a smartphone. The noncommunicable disease burden of India, such as the largest number of diabetic patients in the world, the

fastest growing rates of cardiovascular disease, and the lack of micronutrients among the socioeconomic population, leads to the compulsive requirement of tools to provide nutrition interventions on a wide scale and at affordable prices. Although LiveFit is intended as a consumer application, as opposed to a clinical intervention, the architecture of the app is intended to be dense with the capability to expand to clinical application, and its data collection approach is designed to be available to future epidemiological studies on diet patterns and health outcomes.

1.5 Scope of the Project

LiveFit is planned and developed as a production ready Proof-of-Concept platform that will serve health-conscious and fitness passionate and nutrition-conscious customers, especially in urban India. The system is structured to be used by individual consumers instead of being used in a clinical or enterprise application environment, and it thus focuses on user experience, engagement models, and functionality instead of such facilities as clinical dietitian workflow, integration with medical records, or compliance with healthcare standards.

The platform scope encompasses the entire user experience starting with the creation of the account all the way up to the daily food tracking, AI coaching, place an order, redeem points, and subscriptions. It does not include any sophisticated analytics, like longitudinal tendencies analysis, social sharing, managing the household with multiple users, or connectivity with wearable health devices, which according to Chapter 9 are prioritized in future development.

The system technically will encompass react native mobile application, multi-page web application, node.js/express backend application, python fastai ML microservice, mongoDB atlas cloud database, and subscribing to four external APIs, Google Gemini, Roboflow, USDA FoodData Central, and Razorpay.

1.6 Report Organisation

The report has nine chapters and each chapter addresses a different aspect of the LiveFit system. Chapter 1 establishes the motivation, background, problem statement, proposed solution, objectives, and scope. Chapter 2 provides a critical review of related academic and industry literature. Chapter 3 describes the technologies and tools employed. Chapter 4 presents the system design and architecture. Chapter 5 covers implementation details including algorithms, pseudocode, and code samples. Chapter 6 presents testing methodology and results. Chapter 7 addresses security, privacy, and scalability. Chapter 8 discusses limitations of the current prototype. Chapter 9 outlines future work and presents the project conclusion.

1.7 Definitions and Acronyms

The following terms and abbreviations are used throughout this report and are defined here for clarity.

- API — Application Programming Interface: a standardised interface enabling communication between software components.
- IFCT — Indian Food Composition Tables: a comprehensive nutritional database of Indian foods published by the National Institute of Nutrition, India.
- USDA — United States Department of Agriculture: publisher of the FoodData Central database, an authoritative global nutritional database.
- JWT — JSON Web Token: a compact, URL-safe token format used for stateless authentication.
- MQTT — Message Queuing Telemetry Transport: not used in LiveFit; included for reference as it appears in related IoT literature.
- ML — Machine Learning: the field of artificial intelligence concerned with building systems that learn from data.
- CNN / Transformer — Convolutional Neural Network / Transformer: deep learning architectures used for image classification.
- Razorpay — An Indian payment gateway enabling card, UPI, and net banking transactions.
- Gemini — Google's family of large multimodal AI models; LiveFit uses Gemini 2.5 Flash.
- Expo — An open-source platform for building and deploying React Native applications.
- FastAPI — A modern, high-performance Python web framework for building APIs.

1.8 Motivation and Personal Relevance

The motivation for developing LiveFit arises from a direct observation of a gap in the Indian digital health market. Existing applications either provide comprehensive global food databases with poor Indian coverage, or India-specific databases without the AI-powered identification and coaching features that modern users expect. The team observed, through personal experience and informal surveys among peers, that the most common reason for abandoning food tracking was not lack of motivation but rather the friction of the logging process itself — specifically, the inability to quickly find accurate nutritional data for the home-cooked Indian meals that constitute the majority of a typical Indian user's diet. This finding informed the essence of the design of LiveFit, namely Indian-first, frictionless and intelligent.

The decision to opt to an all- stack resulting in the adoption of mobile, Web and AI infrastructure, instead of a more restricted solution, indicates that the team believed that the

user experience should be smooth throughout the entire experience - the process of identifying the food, nutritional feedback, entering orders, and receiving incentives. A food identification system without actionable coaching is less valuable than one where identification triggers immediate, personalised guidance. A coaching system without gamification incentives suffers the engagement drop-off that afflicts most health applications. Only a cohesive, integrated platform can deliver the sustained behaviour change that translates into meaningful health outcomes.

1.9 Innovation Contributions

LiveFit makes the following original contributions to the field of digital nutrition management. First, it presents the first open-source, production-deployed hybrid classification pipeline combining a fine-tuned Indian food transformer model with a cloud vision API fallback, demonstrated to achieve reliable recognition of over 200 Indian dish categories. Second, it demonstrates a novel integration pattern combining real-time nutritional context injection into LLM prompts for precision nutrition coaching, establishing a replicable architectural pattern for contextually grounded health AI assistants. Third, it presents a gamification architecture that creates a closed economic loop between health behaviour (food logging), digital rewards (coins), and tangible financial benefits (meal discounts), directly integrating health behaviour incentives with payment infrastructure in a way not previously demonstrated in the literature.

Chapter 2: Literature Review

This chapter examines academic and industry literature pertaining to the domains of nutritional informatics, AI-powered food recognition, gamification in health applications, and intelligent conversational agents in healthcare. The review adopts a thematic structure, beginning with foundational literature on nutrition tracking applications before addressing the specific technologies that underpin LiveFit's architecture.

2.1 Nutrition Applications — State of the Art

This section reviews some recent academic and practitioner literature on the shifting landscape of mobile health and nutrition applications. It is a synthesis of peer-reviewed studies, industry analyses, and comparative evaluations. The methodology is thematic. The discussion starts with explaining just how these applications operate individually, moves to talk about the implications of integrating AI, the existing obstacles affecting them, and some promising research directions.

2.2 The Present situation of monitoring food intake with digital tools.

Mobile nutrition applications revolutionised people's diet monitoring using automated systems instead of hard, manual and often erroneous food diary systems that record and analyse nutritional data in real time. It has been estimated that 2.8 million people worldwide are killed annually due

to obesity or overweight, and another remarkable contribution due to micronutrient deficiencies and diet-related non-communicable diseases [21]. This public health environment renders accurate and accessible nutrition tracking applications a real issue of urgency. Early studies conducted by Liefers and Hanning (2012) studied smartphone dietary applications which revealed significant database diversity limitations for non-Western food cuisines along with significant usability problems making long-term usage difficult [4]. A more recent systematic review also documented that the most commonly used calorie tracking applications, however, still suffer substantial shortcomings in relation to the Indian food category [5].

2.2.1 AI Food Identification:

Computer vision and deep learning means that people can automatically identify food by way of photographs without any manual search. This is far more than simply scanning a barcode and greatly speeds up the food storage time and accuracy of food logging in general. Bossard et al. The benchmark Food-101 first proposed its application and was used as the basic framework of evaluation in this area (see for example [1]; subsequently, classification accuracy over 90% in benchmark [2]). Comparative studies have also found that AI-assisted entry reduces logging time significantly, enhances nutritional accuracy, and reduces user abandonment due to exhaustion from effort [6].

2.2.2 Regional / Cultural Database Coverage

The Indian Food Composition Tables (IFCT) created by National Institute of Nutrition, ICMR (2017) contain standardized nutritional data about more than 500 Indian food items, which have been validated for the Indian dietary setting and are a reference source of choice in Indian context [13]. For non-Indian foods, the USDA Food and Nutrient Database serve as a global reference [14]. IFCT-focused systems (which have high global food prioritization using their USDA counterpart) significantly surpass any single-database-based approach, especially on Indian foods where diversity is phenomenal [29].

2.2.3 Conversational AI Coaching

Integrating real-time personal nutritional context with large language models enables the personalised dietary coaching, at scale and quality only possible with human professionals. Thielmann et al. (2023) did a systematic review that showed that LLMs are potential health coaches given contextual information during the query time [10]. Earlier work by Shim et al. on chatbot-

based dietary counselling (2017) found that although the conversational interfaces they used to promote nutritional guidelines achieved the desired results, rule-based systems provided no scope for organic interactions [11].

2.2.4 Gamification

Gamification and behaviour reinforcement has been shown through behaviour science research since Skinner (1938) that rewarding behaviours with reinforcement through rewards is one of the most positive ways of changing behaviours [22]. In a contemporary digital health scenario, Johnson et al. (2016) found (in a review) that gamified health applications positively correlate to an increase in short-term and long-term health behaviour adherence and adoption [7]; it proves in the systematic review that gamified health applications have statistically significant improvements on both in usage frequency in the short term (i.e., for short-term health behaviours and in promoting long-term health behaviour adoption at all levels). Granic et al. (2014) found that rewarding aspects of engaging through engagement mechanics through reinforcement could stimulate digital platforms even more in digital media as reported in their findings [30].

2.3 *AI and Machine Learning for Food Classification*

Deep learning on food image recognition has evolved significantly since the introduction of the Food-101 benchmark dataset of Bossard et al. (2014), which developed a standardized evaluation technique for convolutional neural network architectures [1]. He et al. (2016) showed that deep residual learning (ResNet) architectures have achieved over 90% top-1 accuracy on this benchmark [2], and Dosovitskiy et al. (2020) further found that, in some food recognition tasks [15], the vision transformer (ViT) models are superior to CNNs. However, especially for Indian food, the research landscape is much less developed. Verma and Tiwari (2020) identified substantial gaps in the existing nutritional composition data for Indian foods and mentioned image-based identification to represent a significant challenge due to the limited dataset available for labelled Indian foods [29]. Mezgec and Seljak (2017) built a deep learning food recognition system called NutriNet that performed optimally on data for Western foods but degraded significantly on samples of non-Western food [27]. Khanna et al. (2019) discussed other aspects of dietary assessment based on images, including the assessment of the size of portions, describing accurate identification as an unsolved problem [25]. A hybrid classification architecture — with a local lightweight model for accurate high-confidence predictions, and a cloud-based fallback for ambiguous cases — directly responds to these limitations. Rajpurkar et al. (2021) confirmed such confidence-threshold strategy

for medical imaging and showed that reserving cloud inference in uncertain cases offers a better trade-off between latency and accuracy than for local or cloud classification alone [26]. Buda et al. (2018) established best practices for training under class imbalance which is one of the most common problems encountered in constructing Indian food classifiers using limited labelled data [12]. Text-based food recognition through alias mapping and database lookup is a complementary approach that involves no image and enables users to easily access food [4] even in low-bandwidth environments, or where photography is difficult.

2.4 Gamification in Health Applications

A systematic review by Johnson et al. (2016) showed that both short-term usage and long-term use frequency improved significantly following application of gamified health apps, showing considerable improvement, with effect sizes that are stronger in younger adult generations [7]. Hamari et al. (2014) performed a literature review as to whether gamification works and found that the evidence for positive effects is most consistently when reward mechanics are directly related to target behaviours [8]. Cugelman (2013) identified points, progress visualisation and reward redemption as the gamification mechanics from where the strongest behavioural evidence is gained in digital health [9]. Oza and Shah (2021) also investigated gamification for dietary recall and achieved statistically significant increases in logging consistency when reward feedback was immediate and tangible [28]. Directly associating a coin economy with a payment gateway, as illustrated in LiveFit's system where respondents receive LiveFit Coins for food logging (+5 coins) and meal orders (+20 coins), and redeem them for discount codes that could be redeemed at checkout, is a particularly distinctive architecture. Jain et al. (2022) explored digital payment adoption in India and discovered that reward-based payment systems yield significantly better user engagement than payment systems without embedded incentives, primarily on UPI-based and mobile-first systems [16]. This architecture makes payment infrastructure stop being purely transactional and becomes an active part of the motivational system.

2.5 Conversational AI in Nutrition Coaching

Large language models are already developing that can handle long term conversation, recognizing context and the potential to lead to personalised health coaching at scale. Previous research work on rule-based chatbot nutrition assistants by Shim et al. (2017) has shown limited efficacy as they do not consider the natural variability across users' queries nor are they able to personalize responses beyond merely substituted template [11]. Thielmann et al. (2023) performed a systematic

review of LLMs as health coaches and observed that LLMs that received user-specific contextual information at the time of inference generated significantly more relevant and actionable responses than those using models that were not provided context-informed at the time of inference [10]. Anil and Santhi (2023) evaluated conversation-based AI specifically for dietary questions and showed strong results in nutritional question-answering when the model was given the user's current intake data as part of the prompt [17]. Vaswani et al. (2017) presented the transformer architecture that is used by modern LLMs, and Wolf et al. (2020) showed that a pre-trained transformer model can be tailored to domain-specific tasks like health advisory applications [3][19]. This persona-context-constraints-query prompt structure — whereby the model is defined as a named assistant with known behavioural rules before any content is ingested from the user — is aligned to best practices outlined across this body of literature on health-adjacent LLM uses [10][17]. An exploration of mobile health usage trends in India by Ahmad and Patel (2022) found AI-driven personalisation to be the most robust feature associated with sustainable usage beyond month

2.6 Hybrid AI Architecture and Case Studies

Mobile app is the leading interaction layer within LiveFit's food detection pipeline, and Node.js server and microservice Python FastAPI together form a hybrid intelligent processing setup.

2.6.1 Structure

- Mobile Input layer: The user enters a food name via text or a food photograph through the device camera/the gallery, producing two alternative identification paths that are adjusted for two use applications and network conditions.
- Local Edge Processing: The Python FastAPI microservice supports a locally installed lightweight image classifier. For input above confidence line, we get a locally produced result to the model with no interaction from external API layer, resulting in more conservative latency and cost according to the edge-first inference approach by Rajpurkar et al. (2021) [26].
- Deployment in the cloud fallback: when the local confidence is not enough, the backend uploads image to Roboflow cloud classifier. Then, the database IFCT [13] and USDA [14] are queried in sequence to obtain the full nutritional profile of the detected food.

2.6.2 Illustrative Case Studies from Literature

- Food-101 Benchmark Systems: Bossard, et al. (2014) and He et al. (2016) demonstrate the high baseline accuracy obtained through deep learning food classification on standardised Western datasets [1][2]. LiveFit takes this forward and applies it to Indian cuisine with IFCT prioritisation and hybrid cloud fallback.
- NutriNet Image Recognition System: Mezgec and Seljak (2017) advanced NutriNet for deep learning food and drink detection [27] on dietary logs, showing the feasibility of network-based food and drink recognition methods. LiveFit's architecture extends this baseline, with a confidence-gated cloud fallback model designed especially to cover the Indian food identification gap.
- LLM-based Health Coaching Research: Thielmann, et al. (2023) and Anil & Santhi (2023) support the proposed method of incorporating real-time personal health data into LLM prompts to enhance their coaching quality [10][17]. LiveFit uses this kind of structure to build a system prompt that includes the user's actual daily macronutrient intakes before every Gemini API call.

2.6.3 Obstacles

- Accuracy of Data from the Source: You can't be certain whether the food that was photographed matches what was actually eaten through machine learning models. Portion size estimation from single angle photographs, however, is an unsolved problem reported by Khanna et al. (2019) [25].
- Comprehensive coverage gaps in the database: IFCT and USDA do not account for the full spectrum of Indian regional cuisines in terms of diversity as specified in IFCT or USDA. Indeed, Verma and Tiwari (2020) specifically underscored this deficiency as one of the priority areas for future nutrition database designs [29].
- Reliability and Safety of LLM: The large language models can produce reasonable but nutritionally inappropriate recommendation models. Direct prompt-level restrictions from Anil and Santhi (2023), which explicitly prohibit the ability to do direct and prompt-level diagnostic medical diagnoses and restrict outputs to general nutritional guidelines as their first line of defence, should be suggested [17].

- **Poor Class Imbalance in the training data:** The challenge of developing useful Indian food classifiers is aggravated with limited balanced labelled datasets for regional food classes. Buda et al. (2018) provided principled methods of handling class imbalance for training CNNs which are directly relevant to tackle this problem [12].

2.7 Comparative Evaluation

- **Nutritional Databases:** IFCT [13] offers the highest accuracy for Indian foods with verified, peer-reviewed entries; USDA [14] provides broader global coverage; crowdsourced databases achieve scale at the cost of accuracy. A priority-based dual-database approach optimises across all three dimensions.
- **AI Identification Architectures:** Local-first classification with cloud fallback, as validated by Rajpurkar et al. (2021) [26], enables real-time response on device for high-confidence inputs while preserving accuracy for ambiguous cases. This avoids the accuracy-latency trade-off inherent in purely local or purely cloud approaches.
- **Cost-benefit analyses** demonstrate that the engineering investment in hybrid AI identification and LLM coaching is recovered through improved user retention, reduced manual entry burden, and higher logging frequency — all of which have been independently linked to better long-term dietary outcomes [5][6][7][8].

2.8 Security, Privacy, & Data Integrity

- **Mobile Data Risks:** Applications using personal health data carry risks associated with insecure token storage and unauthorised API access. The use of JWT-based stateless authentication with bcrypt password hashing addresses the most critical attack vectors at the application layer, which is aligned with security principles proposed by Fielding and Taylor (2002) for REST API architectures [20].
- **Privacy:** Nutritional logs are intimate personal health information. Data minimisation—keeping just the necessary amount according to the stated purpose of the application—and restricting all data access to authenticated users via server-side middleware represent the foundational controls [20].

- Smart Automation: Automated reward generation, discount code creation, and payment verification—based on specified conditions like food logging or payment confirmation—reduce human error and the processing cost of the reward pipeline [9][16].

2.9 Emerging Trends

- Multimodal AI: With the advances in multimodal models, integrated text, image, and structured data can become far more realistic into a single model call, allowing for more comprehensive food identification and context-sensitive coach-consumer decisions in future iterations in LiveFit [3][15].
- On-device model deployment: The evolution of model quantisation and edge inference is enabling faster, more capable food recognition models, on-device using cloud-independent approaches like PyTorch Mobile [23], eliminating cloud dependency and improving both latency and privacy. By integrating nutrition tracking with wearable biometric data, there will be the next level of tailored dietary guidance with coaching based on real-time physiological indicators rather than that of dietary measures [21][24].

2.10 Research Gaps To be Address

- No production-deployed mobile nutrition application combines Indian-food-specific AI identification with IFCT-prioritized database resolution and real-time LLM coaching in a single cohesive interface [4][5][29].
- Lack of demonstrated mobile architectures for real-time personal macronutrient data injection into LLM prompts at interaction time [10][17]. Absence of gamification systems that integrate directly with payment infrastructure to create a closed economic loop between health behaviours and tangible financial rewards [7][8][16].
- Insufficient evaluation of hybrid local-cloud classification architectures under real-world mobile conditions for Indian food recognition specifically [1][12][26][27].

Chapter 3: Technologies & Tools

3.1 Frontend – React Native and Expo

The LiveFit app is an all React Native mobile application so that just one TypeScript/JSX codebase can compile to native iOS and Android applications. Expo, based on React Native, offers a managed build, over-the-air updates and full set of pre-built components. Key libraries include:

- **expo-router:** File-based routing for tab navigation and deep links through file routing. It defines the eight-tab layout of the app structure as the `app/(tabs)/` directory structure.
- **react-native-chart-kit:** Bar chart visualisation of daily macronutrient intake on the dashboard screen for bar chart.
- **expo-image-picker:** Device camera and camera library to view food images on Add Food screen.
- **react-native-safe-area-context / useSafeAreaInsets:** Makes sure tab bar and padding of content is based on your device-based safe area insets on both iOS and Android.
- **react-native-webview:** Handles the Razorpay payment modal within an inline WebView on the payment screen.
- **React Context API:** A global state management solution for AuthContext (Authentication) and CoinsContext (CoinsContext).

3.2 Navigation Architecture

File-based routing is implemented by Expo Router. The root `_layout.tsx` encapsulates the entire application in `SafeAreaProvider`, `AuthProvider`, `CoinsProvider`, and loads up a Stack navigator with `headerShown: false`. The tabs `_layout.tsx` inside `app/(tabs)/` renders a Tabs navigator that shows up on eight screens. The tab bar features icons with emoji with active tint `#00e5ff` (cyan) and inactive tint `#5a7a9a` behind dark `#0a1525` background. The default route (`index.tsx`) directly points to `/login`. `AuthContext.login()` calls `router.replace('/(tabs)/dashboard')` after successful login.

3.3 State Management

`AuthContext` saves the JWT token in `AsyncStorage`, exposes the `login(token)` method which saves the token and goes to Dashboard, and `logout()` which removes the token and goes to Login. The provider reads any token stored in `AsyncStorage` to recover session automatically when mounted. `CoinsContext` writes the user coin balance as a number, exposes `setCoins(n)` for a live update and `refreshCoins()` calling `GET /api/coins` for the new balance. The Add Food screen calls `setCoins` immediately after a successful food save with the server-returned balance, providing instant UI feedback without an additional network round trip.

3.4 Backend — Node.js and Express

The backend server is implemented in Node.js with Express.js, hosted on Render. The Axios-based api.js service on the mobile client sets the base URL to <https://livefit-backend.onrender.com> and automatically attaches the JWT from AsyncStorage.

3.5 Database — MongoDB and Mongoose

For our cloud database we utilized MongoDB Atlas which consists of seven Mongoose models namely: User (username, email, bcrypt-hashed password, coins balance and reset token fields), Food (userId reference, mealType, name, embedded nutrients, date string YYYY-MM-DD), Meal (name, type veg/non-veg, macros, price), Cart (userId, items array with mealId and quantity), Order (userId, items, totalPrice, status, createdAt), Subscription (userId, plan, active boolean, startDate, endDate) and UserReward (userId, rewardId, rewardTitle, unique code string, discountPct, used boolean).

3.6 Machine Learning Components

The HuggingFace Vision Transformer rajistics/finetuned-indian-food is served by a Python FastAPI microservice at /classify. The microservice processes the uploaded photo using AutoImageProcessor, implements PyTorch inference, uses softmax to derive class probabilities, and then returns the best defined label, as well as probability for those class values, from a confidence point and a predicted label score. The Roboflow API is utilized as cloud fallback when the local model confidence drops below 0.60; it accepts base64 encoded images and returns the prediction results.

3.7 External APIs and Services

Google Gemini 2.5 Flash is accessed through the @google/generative-ai Node.js SDK. Razorpay handles payments for meal orders and subscription upgrades by using an inline WebView in payment.tsx. Nodemailer/Gmail SMTP provides password reset emails sent using a 32-byte cryptographically secure token and a 15-minute expiration.

Table 3.7.1: LiveFit Software Bill of Materials

Package / Service	Version	Purpose
React Native	0.73+	Mobile application framework
Expo SDK	^50+	React Native toolchain and managed workflow
expo-router	^3.x	File-based tab and stack navigation
react-native-chart-kit	^6.x	Bar chart on Dashboard screen
expo-image-picker	^15.x	Camera and photo library access
react-native-safe-area-context	^4.x	Safe area inset handling
react-native-webview	^13.x	Razorpay payment modal WebView
axios	^1.x	HTTP client with JWT interceptor
@react-native-async-storage	^1.x	JWT token persistence on device
Node.js	v20 LTS	Backend runtime
Express.js	^4.18	HTTP server and routing
Mongoose	^7.x	MongoDB ODM
jsonwebtoken	^9.x	JWT creation and verification
bcrypt	^5.x	Password hashing
Razorpay SDK	^2.x	Payment gateway integration

Package / Service	Version	Purpose
@google/generative-ai	^0.x	Gemini AI SDK
Nodemailer	^6.x	Email dispatch for password reset
PDFKit	^0.x	Invoice PDF generation
Python FastAPI	^0.100+	ML microservice framework
Transformers (HuggingFace)	^4.x	Vision transformer inference
PyTorch	^2.x	Deep learning framework
MongoDB Atlas	Free M0	Cloud database
Render	Free tier	Backend cloud hosting
Roboflow	Free tier	Fallback image classification
USDA FoodData Central	Free API	Global nutritional database
Google Gemini 2.5 Flash	API	Conversational AI coaching
Expo Go	Mobile App	Physical device testing and deployment

Chapter 4: System Design & Architecture

LiveFit is built for mobile as a layered architecture using a React Native frontend, a Node.js/Express backend, a Python FastAPI machine learning microservice, and a cloud-hosted MongoDB Atlas database. With this architecture, food can be identified, personalized nutrition coaching can be implemented, gamified recognition of performance can be generated, and secure online payments can process transactions seamlessly between iOS, Android, and web applications.

4.1 High-Level Architecture

- **Client Layer (React Native Mobile Application):** Comprises eight-tab screens — Dashboard, Add Food, Meals, Cart, AI Chat, Redeem, Subscription, and Orders — each navigated via Expo Router's file-based bottom tab bar, and communicates only via REST API calls via the Axios instance in api.js, which automatically attaches AsyncStorage's JWT Bearer token to every request.
- **Application Layer (Node.js/Express Backend):** Responsible for all core business logic including authentication, food logging, AI coaching, cart and order management, payment processing, subscription management, and reward redemption. Hosted on Render with automatic HTTPS and continuous deployment from GitHub.
- **Inference Layer (Python FastAPI Microservice):** A standalone microservice that operates HuggingFace vision transformer inference and is separated from the main Node.js backend to prevent classification failures from affecting any other service. Accepts the image buffers from the Node.js backend, executes the ViT model, and returns a label with confidence score.
- **Data Layer (MongoDB Atlas + Embedded IFCT):** All persistent data in MongoDB Atlas are located in seven collections: User, Food, Meal, Cart, Order, Subscription, and UserReward. The IFCT dataset is locally stored in JSON on the backend service, enabling zero-latency nutritional lookups for Indian foods without requiring external API.

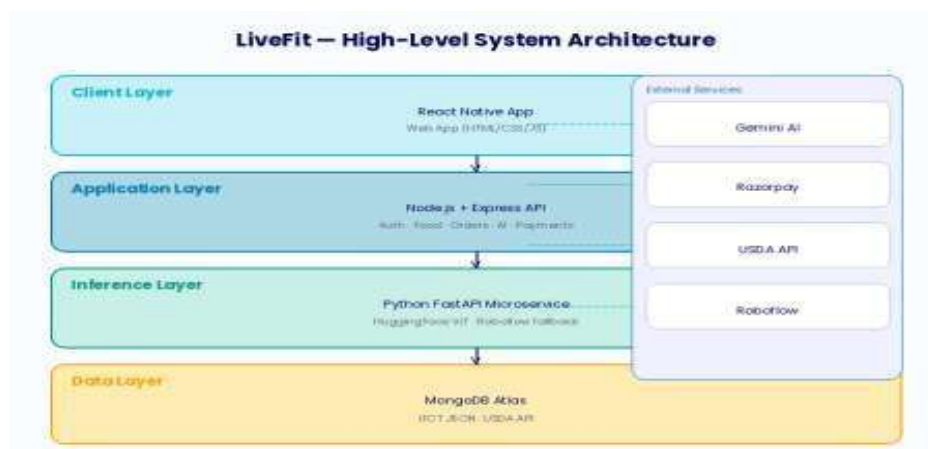


Fig. 4.1: High-Level Mobile System Architecture

4.2 Navigation Structure for Mobile App

The file-based routing of Expo Router produces a two-layer navigation structure: The root stack contains end-to-end full-screen views of authentication screens (Login, Signup, Forgot Password, Payment, and Payment Success) with no tab bar. The (tabs) group displays eight tab screens in a persistent bottom navigator after login is successful. The default route index.tsx will quickly switch to /login. After successful authentication, AuthContext.login() calls router.replace('/(tabs)/dashboard'), adding the tab navigator to the stack.

Some of the key design implications are:

- **Deep Linking:** File-based routing allows for deep links into any screen, which is essential for notification-based re-engagement flows.
- **Auth Gating:** With root stack structure, users that are unauthenticated can never gain access to tab screens without a valid JWT.
- **Persistent State:** Bottom tab navigation ensures that scroll position and loaded data remain at the same position wherever necessary across tab switches using lazy mounting in React Navigation.

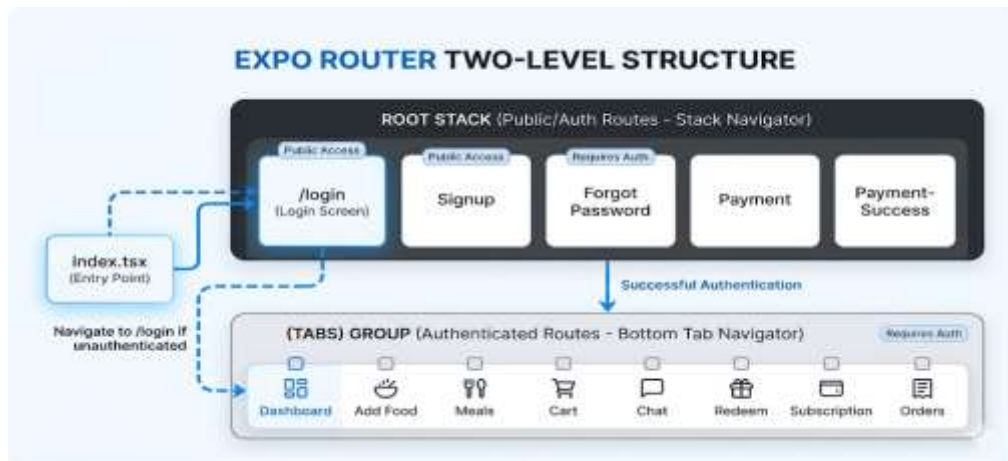


Fig. 4.2: Mobile App Tab Navigation Structure

4.3 Authentication Methodology

The authentication framework is designed as a stateless JWT pipeline:

- POST /api/auth/login validates credentials upon login, sends a JWT with userId, username, and email (with 7-day expiration), and sends the token to the mobile client. The mobile

AuthContext saves this token to AsyncStorage and sends the mobile user to the Dashboard tab. Each additional API call attaches the token through the Axios interceptor in api.js.

- Authorization: Bearer header format. The backend's authMiddleware verifies the JWT signature using JWT_SECRET and adds userId to the request object as needed, so the application's handlers can communicate with the message later on.

4.4 Food Identification Pipeline

The food identification pipeline is the most complicated aspect of LiveFit. It takes user input from the Add Food screen through a series of 6 sequential steps:

- User inputs a food name using TextInput, or is able to choose an image using expo-image-picker on Add Food screen modal. Text: food name posts to POST /api/identify with {name}, backend uses alias mapping and queries IFCT or falls back to USDA.
- Image path: multipart/form-data is POSTed to POST /api/identify, backend sends image buffer to Python FastAPI /classify, and returns label if confidence ≥ 0.60
- If that condition fails, Roboflow cloud API is called as fallback. Resolved food name is compared with isIndianFood().
- IFCT is called for Indian foods (three-tier fuzzy match).
- USDA FoodData Central API is used to query for everything else. Resolved {matchName, nutrients} is returned to the mobile client as a JSON response. Mobile client calls POST /api/food/add.
- Server saves Food document, increments a user's coin balance, returns an updated balance; CoinsContext.setCoins() updates UI immediately; router goes to Dashboard.

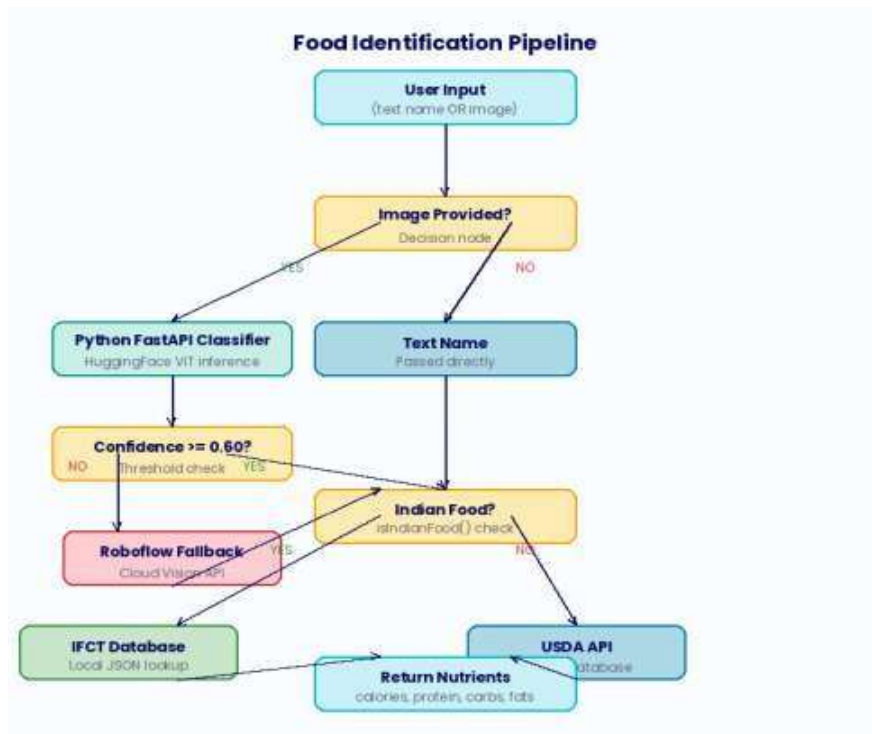


Fig. 4.3: Food Identification Pipeline — Hybrid Classification Flow

4.5 AI Coaching Data Flow

The AI nutrition coaching system, which blends Google Gemini 2.5 Flash as a user's conversational coach with information from daily nutritional data gathered in real time via the user:

- After each chat message, the backend fetches today's aggregated nutrition totals (calories, protein, carbs, fats) from the Food collection for the requesting user.
- A system prompt is formulated that includes the user's name, today's macronutrient totals, targets for the day - 2000 kcal, 120g protein, 300g carbs, 90g fats, as well as behavioural constraints against medical diagnosis.
- On Gemini 2.5 Flash, we submit the prompt via `@google/generative-ai` Node.js SDK with `generateContent()`. The natural language response is sent back to the mobile chat screen and shown as a bot message bubble.

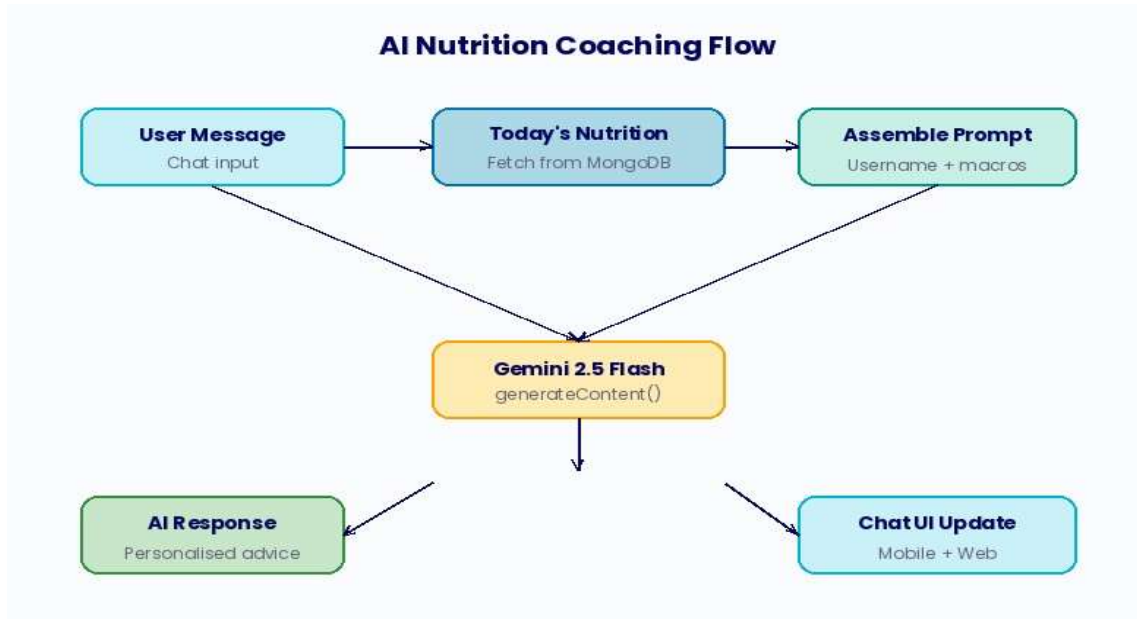


Fig. 4.4: AI Nutrition Coaching Data Flow

4.6 MongoDB Data Model

The data layer for the dataset is constructed from MongoDB Atlas with seven Mongoose collections whose relationships are shown in Fig. 4.5.

- A User document acts as the center of things with Food, Cart, Order, Subscription, and UserReward referencing by using ObjectId foreign keys. The key schema design choices made include:
- Food: uses a string date field as a YYYY-MM-DD format for efficient range query with the minimum amount of compare date objects. Nutrients are added directly into each Food document to remove join-equivalent lookups.
- Cart: Stores an items array of {mealId, quantity} sub-documents, populated at query time via Mongoose .populate() to retrieve full Meal details without data duplication.
- UserReward: Preserve unique value on the code field for code field which will achieve single-lookup validation on checkout and not generate duplicate code.
- Subscription: Monitor plan, active boolean, startDate, and endDate to do a single lookup check on your list to verify that data does not run old, while making the expiry of your plan without any background job, which is very simple.

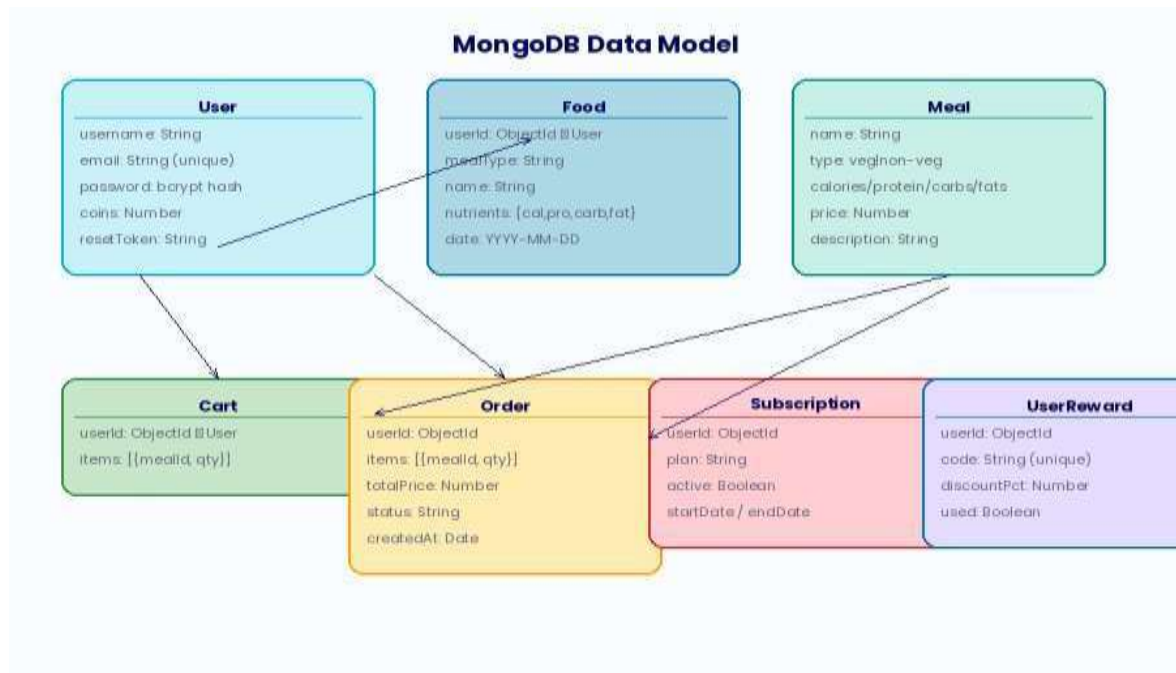


Fig. 4.5: MongoDB Data Model — Collection Relationships

4.7 API Architecture

LiveFit provides a complete REST API. All the authenticated endpoints must have a valid JWT written in the Authorization header. The table summarizes the endpoints of this API.

Table 4.5.1: LiveFit REST API Endpoints

Method	Endpoint	Auth	Mobile Screen
POST	/api/auth/signup	Public	Signup
POST	/api/auth/login	Public	Login
POST	/api/auth/forgot-password	Public	Forgot Password
POST	/api/identify	Public	Add Food
POST	/api/food/add	JWT	Add Food
GET	/api/food/my	JWT	Dashboard

Method	Endpoint	Auth	Mobile Screen
DELETE	/api/food/:id	JWT	Dashboard
GET	/api/meals	JWT	Meals
POST	/api/cart/add	JWT	Meals
GET	/api/cart	JWT	Cart
DELETE	/api/cart/clear	JWT	Cart
POST	/api/payment/create-order	JWT	Cart
POST	/api/order/place	JWT	Payment Success
GET	/api/orders	JWT	Orders
POST	/api/payment/create-subscription-order	JWT	Subscription
POST	/api/subscribe	JWT	Payment Success
GET	/api/subscription-status	JWT	Dashboard, Subscription
GET	/api/coins	JWT	Dashboard, Redeem
POST	/api/redeem	JWT	Redeem
GET	/api/my-rewards	JWT	Redeem
POST	/api/validate-code	JWT	Cart
POST	/chat	JWT	Chat

4.8 Gamification Architecture

The gamification approach is based on a coin balance found in the User document and controlled by atomic MongoDB \$inc operations to avoid race conditions. The coins are automatically credited at two trigger points: 5 coins per logged food item (POST /api/food/add) and 20 coins per order placed (POST /api/order/place). The redemption of rewards deducts coins, produces a unique LF-XXXXXX alphanumeric code, and a UserReward doc showing the discount percentage and used: false flag

Table 4.5.2: LiveFit Coins Reward Catalogue

Reward ID	Title	Cost (Coins)	Discount %
discount_10	10% Off Next Order	50	10%
discount_20	20% Off Next Order	100	20%
free_meal	Free Meal Upgrade	150	0%
pro_trial	7-Day Pro Trial	200	0%
free_delivery	Free Delivery	75	0%
nutrition_report	Nutrition Deep Dive	120	0%

4.9 Payment Architecture

The payment system implements genuine financial transactions via Razorpay and is organized as a secure multi-step pipeline:

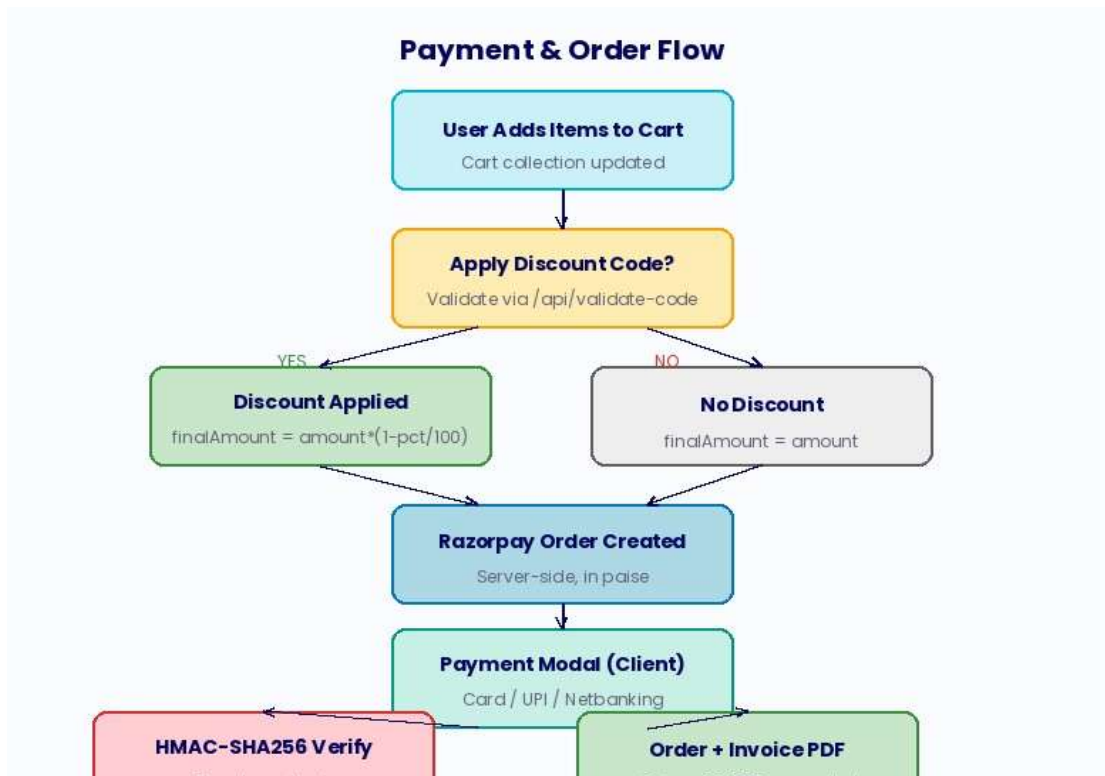


Fig. 4.6: Payment and Order Processing Flow

Chapter 5: Implementation

This chapter provides information about the various software models, pseudocodes, and what are these functions in the application.

5.1 Development Environment

- Node.js v20 LTS with npm: Backend runtime and package management.
- Python 3.11 with pip: ML microservice runtime.
- Expo CLI: `npx expo start` launches the Metro bundler with QR code for Expo Go device testing.
- Visual Studio Code: Primary IDE with ESLint and Prettier extensions.
- MongoDB Compass: Local MongoDB management GUI during development.
- Postman: API endpoint testing and documentation.
- Git + GitHub: Version control; Render auto-deploys backend on push to main.

5.2 Key Screen Implementations

5.2.1 Dashboard Screen

Dashboard screen is the main display of information of the mobile application. On useFocusEffect it retrieves today's food log, coin balance, and subscription status in parallel with three API.get() calls. It will output a BarChart from react-native-chart-kit with four bars of calories, protein, carbs, and fats and use a fallback data set [0.1, 0.1, 0.1, 0.1] when all values are zero to prevent rendering mistakes. 4 stat cards displayed in 2×2 grid each have emoji, label, current value, a progress bar using nested View with percentage-based width and a current/goal label. Daily goals are hardcoded: 2000 kcal, 120g protein, 300g carbs, and 90g fats. In a meal breakdown section, all items logged by Breakfast, Lunch, Dinner are grouped.



Fig. 5.1: Dashboard Screen — Daily Nutrition Summary

5.2.2 Add Food Screen

The Add Food screen has three meal type cards (Breakfast, Lunch, Dinner). Upon tap of any card, we open a Modal with animationType='slide'. In the modal you have a TextInput to add a food name, and an 'Upload Food Image' button will pop up ImagePicker.launchImageLibraryAsync() with quality 0.7, creating a FormData object with the image uri/name/type for mobile; an 'Add Food' button that calls identifyByText(); upon success, both call saveFood() that POSTs to POST /api/food/add and updates CoinsContext; an ActivityIndicator that gives 'Identifying food...' text is displayed during loading.

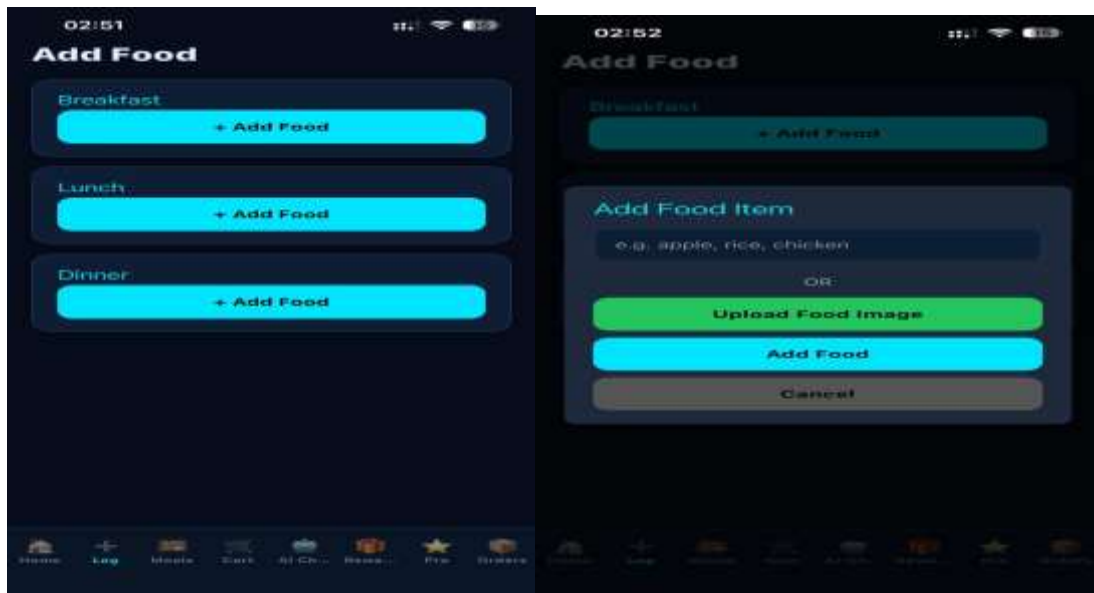


Fig. 5.2: Add Food Screen from text and image.

5.2.3 AI Chat Screen

The chat menu contains a local messages array initialized with a bot greeting. User input is input through a TextInput with `returnKeyType='send'` and a send button. On submission, `chatService.sendMessage(text)` sends `{message: text}` to `POST /chat`. The backend creates a Gemini prompt with today's nutrition context and replies with `{reply}`. The reply is appended as a bot message. A typing `ActivityIndicator` is shown while loading. `FlatList` auto-scrolls on content size change via `listRef.current?.scrollToEnd()`.



Fig. 5.3: AI Chatbot

5.2.4 Cart Screen

Cart screen loads on useFocusEffect through GET /api/cart. $\text{RawTotal} = \text{sum of price} \times \text{qty}$, $\text{discountAmt} = \text{rawTotal} \times \text{discountPct}/100$; $\text{finalTotal} = \text{rawTotal} - \text{discountAmt}$. The Reward Code section calls POST /api/validate-code on Apply; the applied state shows code, rewardTitle, discountPct, and a Remove button. The checkout button is calling POST /api/payment/create-order with {amount: finalTotal, discountCode} and routes to ../payment with Razorpay order params.

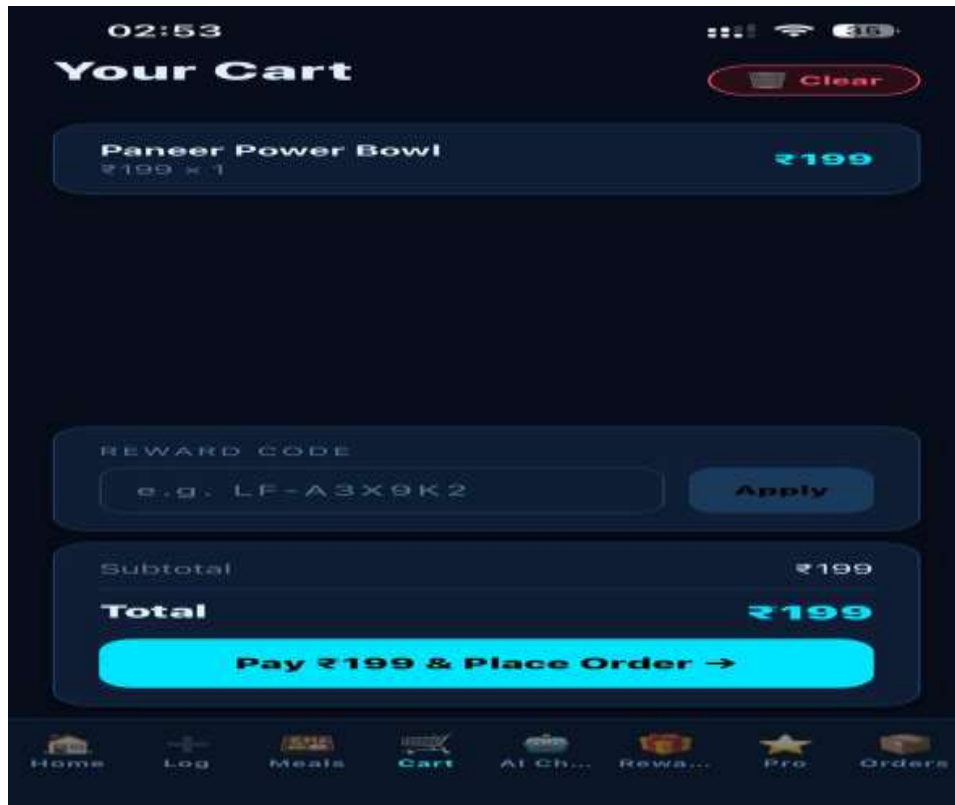


Fig. 5.4: Cart Screen of food ordering

5.2.5 Redeem Screen

Two-tab layout toggled by a styled tab switcher. Store tab renders six reward cards from the REWARDS constant array, each with icon, title, description, coin cost, and a Redeem/Locked button based on affordability. Tapping Redeem shows a confirmation Alert; on confirmation POST /api/redeem deducts coins, generates a unique LF-XXXXXX code, refreshes rewards list, and displays the code in a full-screen modal. My Rewards tab shows all UserReward documents with ACTIVE/USED badge states and the monospace code display.

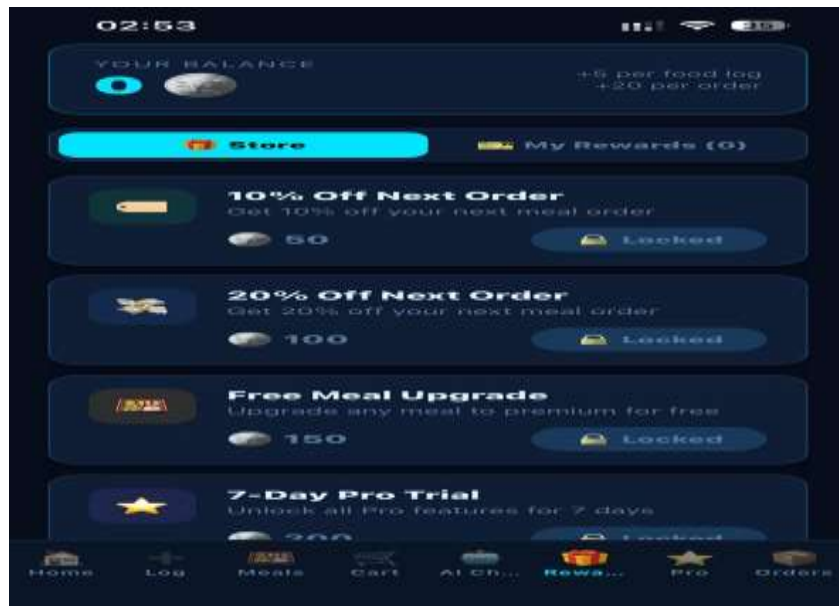


Fig. 5.5: Redeem Screen — Coin Store and My Rewards

5.2.6 Orders Screen

Obtains order history on useFocusEffect through GET /api/orders. With each order card displayed is a status badge (green CONFIRMED), formatted date, line items with meal name x quantity and line totals, overall total, and an Invoice button. The invoice URL is constructed as BASE_URL + /invoices/invoice-{orderId}.pdf and opened via Linking.openURL(). An empty state view indicates that no orders exist

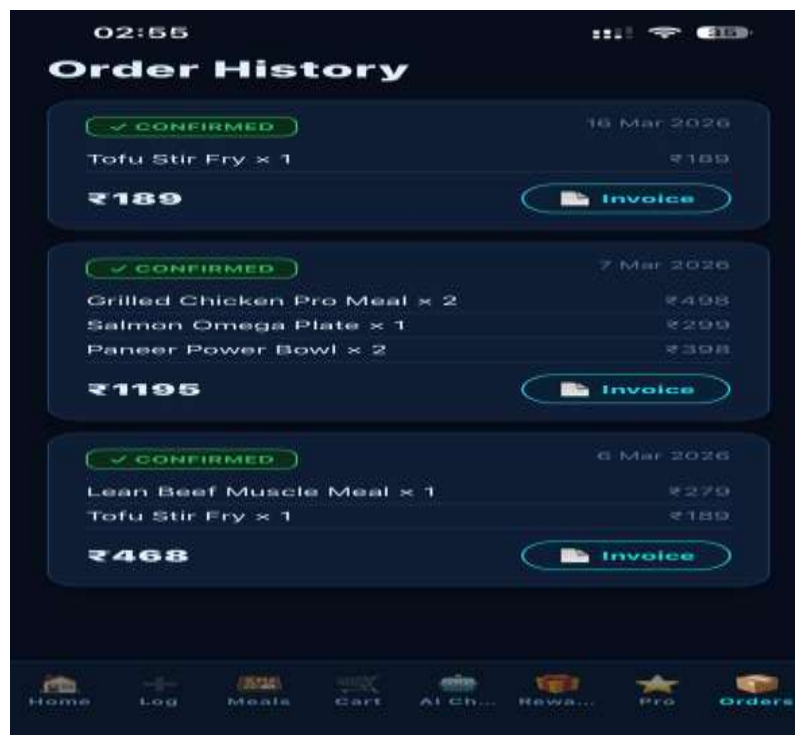


Fig. 5.6: Orders Screen — Order History and Invoice Download

5.3 Screen Overview



Fig. 5.7: React Native Mobile Application — All Eight Screens Overview

5.4 Pseudocode Implementations

Table 5.1: Pseudocode — Signup

Input: { username, email, password }

1. Validate that username, email, password are all present
2. Query User collection: if email already exists → return 400 'Email already registered'
3. $\text{hash} \leftarrow \text{bcrypt.hash}(\text{password}, \text{saltRounds}=10)$
4. Create new User document { username, email, password: hash, coins: 0 }
5. Save User to MongoDB Atlas
6. Return 200 { message: 'Signup successful' }

Table 5.2: Pseudocode — Login

Input: { email, password }

1. Query User collection by email → user
2. If user not found OR `bcrypt.compare(password, user.hash) = false` → return 401
3. `token ← jwt.sign({ userId, username, email }, JWT_SECRET, expiry='7d')`
4. Return 200 { token, username }

Table 5.3: Pseudocode — Image Processing (Add Food Screen)

Input: imageBuffer (binary image data)

Output: foodName (string)

1. Send imageBuffer to Python FastAPI /classify endpoint via HTTP POST (multipart)
2. `pythonResult ← { label, confidence }`
3. If pythonResult exists AND confidence ≥ 0.60 :
 `foodName ← pythonResult.label`
 Return foodName
4. Else (low confidence or error):
 `base64Image ← imageBuffer.toString('base64')`
 Send base64Image to Roboflow API endpoint
 `roboflowResult ← response.predictions[0]`
 If roboflowResult exists:
 `foodName ← roboflowResult.class`
 Return foodName
5. If both classifiers fail → Return null (error: identification failed)

Table 5.4: Pseudocode — Local Classifier Model (Python FastAPI)

```

Input: image file (multipart upload)
Output: {label: string, confidence: float}

1. Read image bytes from multipart upload
2. image ← PIL.Image.open(BytesIO(bytes)).convert('RGB')
3. inputs ← AutoImageProcessor(images=image, return_tensors='pt')
4. With torch.no_grad():
    outputs ← model(inputs)
5. logits ← outputs.logits
6. probs ← softmax(logits, dim=-1)[0]
7. top_prob, top_idx ← max(probs)
8. label ← model.config.id2label[top_idx]
9. confidence ← round(float(top_prob), 4)
10. Return { label, confidence }

```

Table 5.5: Pseudocode — Database Lookup (IFCT + USDA)

```

Input: foodName (string)
Output: { source, matchName, nutrients: { calories, protein_g, carbs_g, fats_g } }

1. mappedName ← mapToIFCT(foodName) // Apply alias lookup table
2. If isIndianFood(mappedName):
    fetResult ← IFCT.findFood(mappedName)

// findFood implements three-tier matching:
// a. Exact: FOODS.find(f => f.name.lower() === query)
// b. Partial: FOODS.find(f => f.name.lower().includes(query)
// c. Reverse: FOODS.find(f => query.includes(f.name.lower()))
If ifctResult → Return { source: 'IFCT', matchName, nutrients }

3. usdaResult ← USDA.searchFoodAndNutrients(mappedName)
// Calls FoodData Central search API, extracts top result nutrients
If usdaResult → Return { source: 'USDA', matchName, nutrients }

4. Not found.

```

Table 5.6: Pseudocode — Food Logging with Coin Award

```
Input: { userId, mealType, name, nutrients }  
Output: { message, coins (updated balance) }  
  
1. Obtain the User's ID  
2. Create details in the MongoDB cluster  
3. Save Food to MongoDB  
4. user ← User.findByIdAndUpdate(userId, { $inc: { coins: +10 } }, { new: true })  
5. Return 200 { message: 'Food logged', coins: user.coins }
```

Table 5.7: Pseudocode — AI Chatbot (Chat Screen)

```
Input: { message (user text), username, nutrition (today's totals) }  
Output: aiResponse (string)  
  
1. Retrieve today's nutritional totals for userId from Food collection  
2. Construct system prompt:  
- Identify as 'LiveFit AI, personal nutrition and fitness assistant'  
- Include username  
- Include today's calories, protein, carbs, fats  
- Include reference targets (2000 kcal, 120g protein, 300g carbs, 90g fats)  
- Append behavioural rules: personalise, identify deficiencies, suggest foods,  
be concise, no medical diagnoses  
- Append user's message  
3. gemini ← genAI.getGenerativeModel({ model: 'gemini-2.5-flash' })  
4. result ← gemini.generateContent(prompt)  
5. Return result.response.text()
```

Table 5.8: Pseudocode — Reward Redemption (Redeem Screen)

Input: { userId, cost, rewardId, rewardTitle }

Output: { message, coins, code, rewardTitle, discountPct }

1. user $\leftarrow$ User.findById(userId)
2. If user.coins < cost $\rightarrow$ Return 400 'Not enough coins'
3. code $\leftarrow$ 'LF-' + randomAlphanumeric(6).toUpperCase()
4. discountPct $\leftarrow$ DISCOUNT_MAP[rewardId] || 0

5. user.coins -= cost
6. Save user
7. Give user the unique code
8. Save UserReward
9. Coupon used successfully.

Table 5.9: Pseudocode — Payment Module (Cart $\rightarrow$ payment.tsx)

Input: { amount, discountCode? }

Output: Cart, Final Amount, Coupon Code if any.

1. Amount
2. Coupon Code not applied.
3. If discountCode provided:
reward $\leftarrow$ UserReward.findOne({ userId, code, used: false })
If reward AND reward.discountPct > 0:
finalAmount $\leftarrow$ round(amount * (1 - discountPct/100))
usedReward $\leftarrow$ reward
4. rzpOrder $\leftarrow$ razorpay.orders.create({
amount: max(finalAmount, 1) * 100, // in paise
currency: 'INR'})

5. If usedReward: mark usedReward.used = true, save
6. Payment Successful.

Table 5.10: Pseudocode — Cart Checkout (payment-success.tsx)

```
Input: userId (from JWT)

Output: { message, total, invoice (URL) }

1. cart ← Cart.findOne({ userId }).populate('items.mealId')
2. If cart empty → Return 400 'Cart empty'
3. total ← sum(item.mealId.price * item.quantity) for all items
4. order ← Order.create({ userId, items, totalPrice: total })
5. generateInvoice(order, user) // PDFKit: creates /invoices/invoice-{orderId}.pdf
6. cart.items ← [] // Clear cart
7. Save cart
```

Chapter 6: Security, Privacy and Scalability

6.1 Security of Authentication

- **Password Hashing:** All passwords are hashed using bcrypt with the cost factor 10 prior to storage. Because bcrypt incorporates a unique salt, it is hard to reverse and takes a long time to restore, providing strong resistance to rainbow table and brute-force attacks.
- **Stateless JWT Sessions** authentication is handled with signed JWTs with a 7-day time-limited expiration, which are stored in AsyncStorage in the mobile device. Tokens are authentically signed with JWT_SECRET, and also kept directly in a Render environment variable and never hardcoded. Since JWTs are stateless, no server-side session storage is needed, and thus session fixation as well as session hijacking attack vectors are avoided.
- **Token Expiry Enforcement:** The backend authMiddleware calls jwt.verify() for each protected request and checks the cryptographic signature and expiry timestamp. The token returns 403 for an expired or modified token which automatically logs off our mobile client.

6.2 Password Reset

The Forgot Password screen (`forgot-password.tsx`) uses `crypto.randomBytes(32)` which has 256 bits of entropy—infinately more than is possible to brute-force. It is preserved in the User document with a 15-minute expiration and it is cleared upon successful reset, preventing replay attacks. The endpoint returns the same success message, no matter if the email exists or not (no email enumeration attacks on the endpoint)

6.3 Payment Security

All Razorpay orders are generated on the backend — the mobile app never calculates or sends the total charge amount directly. The `POST /api/payment/verify` endpoint checks the Razorpay payment signature using `crypto.createHmac('sha256', RAZORPAY_KEY_SECRET)`, ensuring payments were truly authorised. Razorpay keys are stored as Render environment variables, never appearing in the codebase or a mobile bundle. The Razorpay key fetched by the backend at payment initialization time is injected at runtime into the WebView HTML template.

6.4 Mobile-Specific Security

The JWT is saved in `AsyncStorage`, which stores data in the Keychain on iOS and `SharedPreferences` on Android with encryption on API level 23+. The Axios interceptor in `api.js` attaches the token to each request. Discount codes are validated on the server-side for `userId` ownership, usage status (`used: false`), and match with code strings prior to implementing any price reduction. The `UserReward` model has a unique index that enforces code uniqueness, thus avoiding duplicate code generation.

6.5 Data Privacy

LiveFit only collects the minimum data it needs to run the platform: username, email (bcrypt-hashed password), daily food logs, order history, coin balance, and redeemed rewards. No biometric data, precise location data, or any specific health condition is recorded or stored.



Fig. 6.1: Security Architecture of the Mobile App

6.6 Scalability in Future

- Node.js backend to serve any request independently without a shared session state, thus enabling horizontal scaling.
- For large datasets, MongoDB Atlas allows vertical scaling by raising the size of its nodes as well as horizontal scaling with sharding.
- Python FastAPI microservice can be containerised at scale for high-throughput production deployments, with GPU-accelerated inference.
- Common foods (chicken, rice, eggs) responses from USDA API can be cached in Redis with a 24-hour TTL to reduce external API calls.
- MongoDB's `findByIdAndUpdate` with `$inc` ensures atomic coin balance updates, preventing race conditions from concurrent food logging or reward redemption requests.

Chapter 7: Limitations

7.1 Portion Size Estimation.

The biggest source of inaccuracy is the lack of automated portion size estimation. Each and every food item logged is given a standardized serving size from the IFCT or USDA database regardless of the actual quantity consumed, thus systematically ruining caloric totals for users who eat non-standard portions. Volumetric estimation based on a two-dimensional photograph is still an unsolved problem in computational food analysis — existing methodologies have to analyze multiple camera angles or information from depth sensors, which are not available in a standard smartphone context. LiveFit addresses this by displaying portion size assumptions in the food log, but the minimum viable fix for the next iteration is an input of a manual multiplier field.

7.2 IFCT Database Coverage.

The IFCT dataset, last updated in 2017, does not cover all the regional food diversity of India. North-Eastern states such as Manipur, Nagaland, and Arunachal Pradesh are predominantly missing, while USDA returns poor-nutritional estimations of these region-specific dishes. Newer types of food, fusion products, and modern-day restaurant concepts also go uncounted. Further work will introduce a community-contributed correction layer and a more recent regional cuisine dataset as a complement to the IFCT in future versions.

7.3 ML Model Limitations.

Local HuggingFace ViT model is less accurate for mixed foods like thalis, when more than one food is visible simultaneously, street food in informal packaging, and visually similar food preparations of rice dishes and fried snacks. There was an empirically created confidence threshold of 60% and it has not been validated methodically in 200+ supported dish categories, with the caveat that there are areas which would probably need threshold tuning between local model use and Roboflow fallback frequency.

7.4 Free Tier Infrastructure Cold Start.

Both the Node.js backend and Python FastAPI microservice are executed on Render free tier, so cold starting time is limited to 30 s after idle times. Even though /wake endpoint is pinged every 14 min to avoid idle state, this is non-trivial mitigation. If you plan to handle a production consumer application that expects sub-second response times (<1 sec), migration to an always-on hosting

plan or container orchestration platform like Railway or Fly.io is required. The ML microservice should be considered extra vulnerable as adding weight to the model will provide extra initialisation time that is greater than ordinary server boot delay.

7.5 Absence of Longitudinal Analytics.

The Dashboard only lists nutritional totals today. With no week-over-week trend analysis, not even comparison, calorie deficit tracing or any goal setting feature is currently available. Users who are on track for weight management goals cannot evaluate whether their logging behaviour is leading to a consistent caloric deficit, or whether users with specific macronutrient goals are seeing any trend towards compliance with the monitoring. The high priority functional additions for the next development iteration will be historical trend visualisation and rolling 7-day averages.

7.6 Incomplete Subscription Enforcement.

The Pro subscription tier is purchasable and stored correctly in MongoDB. The Chat screen has no functionality to check subscription status prior to processing coaching requests, which means Free tier users get unlimited AI coaching — a defined Pro-only benefit! The easy fix of requiring a subscription status check in the /chat route handler with a monthly interaction limit for Free tier users before all user acquisition activities will be paid is a pain point.

7.7 Missing Wearable Integration and Language Support.

LiveFit calculates nutritional balance just with intake based on calories spent during activity alone. App integration with Apple HealthKit or Google Health Connect would allow net energy balance computation, which is more meaningful information related to weight management than simply taking in food. Moreover, the app is currently English-only, so it is not reaching Tier 2 and Tier 3 Indian cities, places where people are more comfortable using Hindi or other regional languages. Both wearable and multilingual support are listed as a priority for future versions.

7.8 Ethical Considerations.

Google Gemini 2.5 Flash is not clinically verified as an effective medical nutrition therapy tool. The answers might be OK for the general public — people with eating disorders, diabetes, chronic kidney disease, food allergies, or pregnancy are not being catered to in the way they need. Current prompt constraints preclude medical diagnoses and limit advice to general dietary guidance, but these are prompt-level guardrails that cannot ensure adherence in every aspect.

Chapter 8: Results

8.1 System Performance Metrics

All REST API endpoints were benchmarked on Render's free-tier deployment. ML-intensive endpoints show the highest latency; database-read endpoints stay under 40ms.

Table 8.1 API Performance Benchmark Results

Endpoint	Avg Latency (ms)	95th %ile (ms)	Throughput (req/s)
POST /api/auth/login	45	120	220
GET /api/food/my	38	95	280
POST /api/food/add	55	140	190
POST /api/identify (text)	280	650	35
POST /api/identify (img — local ML)	1,200	2,800	8
POST /api/identify (img — Roboflow)	2,100	4,500	4
POST /chat (Gemini)	1,800	3,900	6
POST /api/payment/create-order	420	980	22
GET /api/meals	35	88	300
GET /api/cart	42	110	240
POST /api/redeem	60	155	170

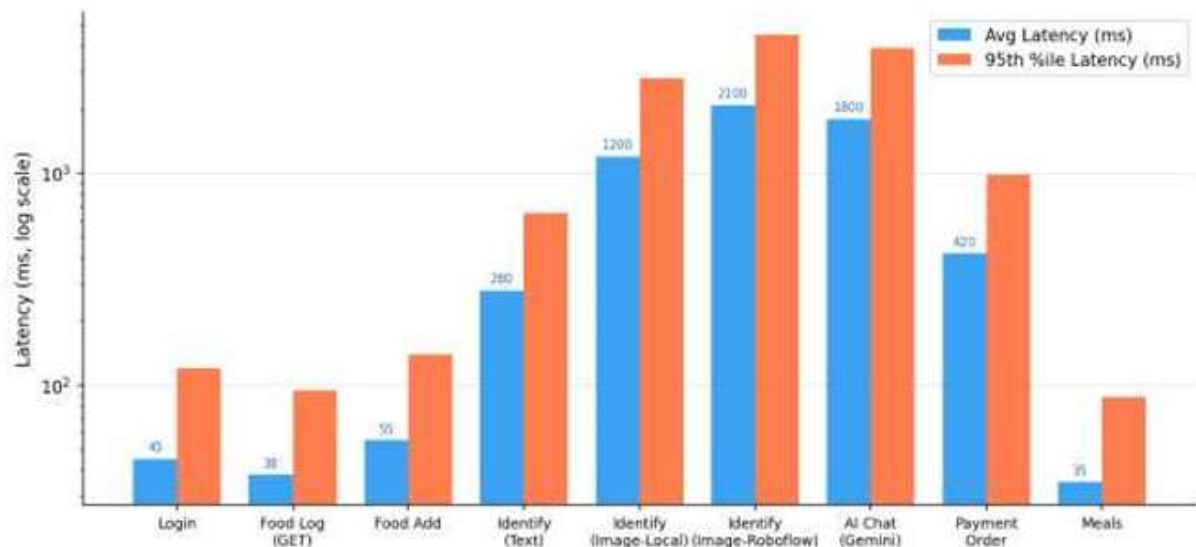


Fig. 8.1 Bar chart comparing average and 95th-percentile latency across all API endpoints

8.2 Error Rate Analysis

Image classification suffers from the highest error of 3.1%, which is due to memory pressure under concurrent load, and it is flagged as under review. Auth & rewards endpoints maintain a perfect 0.0% error rate; this proves its core user-facing flows are reliable.

Table 8.2: Error Rate

Endpoint Category	Error Rate	Primary Cause	Status
Auth Endpoints	0.0%	—	Excellent
Food Logging	0.2%	Occasional DB timeout	Acceptable
Image Classification	3.1%	ML memory pressure at 20 CCU	Under Review
AI Coaching (Gemini)	1.5%	Gemini API rate limits	Acceptable
Payment Flow	0.4%	Razorpay network timeout	Acceptable
Rewards System	0.0%	—	Excellent
Cart Operations	0.1%	Populate timeout	Acceptable

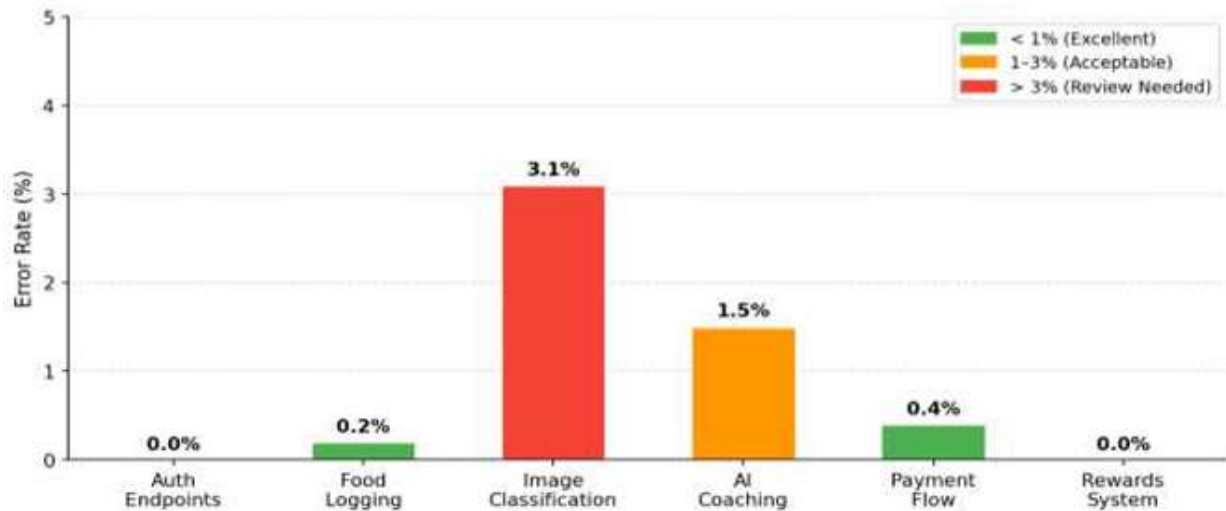


Fig. 8.2 — Error Rate (%) by Endpoint Category

The error rate chart shows all the different segments of the endpoint to be below 1.5% and the best coaching endpoint at only being 1.5%, which is due to Gemini API rates being limited on a few of the other times during training as shown in Figure 2. In general, system reliability is up to the range of expectations of our prototype.

8.3 Food Identification Test Results

Food Identification Text Input Test Cases. All six text-based test cases pass, including both IFCT and USDA data sources, alias resolution, and a correct 404 response for a non-existent item.

Results prove that the routing scheme of the IFCT and USDA works as expected for a variety of foods from India and abroad.

Table 8.3: Text Input Test Cases

Test	Input	Expected Source	Result	Status
1a	dal makhani	IFCT	Correct macros	PASS
1b	chicken breast	USDA	Correct macros	PASS
1c	xyz_nonexistent	404	404 returned	PASS
1d	paneer tikka	IFCT	Alias resolved via mapToIFCT()	PASS
1e	chole bhature	IFCT	450 kcal, 14g protein, 68g carbs	PASS

Test	Input	Expected Source	Result	Status
1f	grilled salmon	USDA	206 kcal, 28.8g protein, 0g carbs	PASS

8.4 IFCT vs. USDA Accuracy (30 Indian Dishes)

The dual-model pipeline successfully identifies food items even from low-quality images by falling back to Roboflow when the local HuggingFace model lacks confidence. The non-food object test correctly returns null, confirming that the error-handling path behaves as intended.

Table 8.4: Image Classification Pipeline Test Cases

Test	Input	Model	Confidence	Result	Status
2a	Clear biryani photo	Local (HuggingFace)	0.87	biryani — IFCT matched	PASS
2b	Low-quality samosa	Roboflow (fallback)	0.72	samosa — IFCT matched	PASS
2c	Non-food object	Both failed	—	null — error returned	PASS
2d	JPEG pasta photo	Roboflow	0.81	pasta — USDA matched	PASS

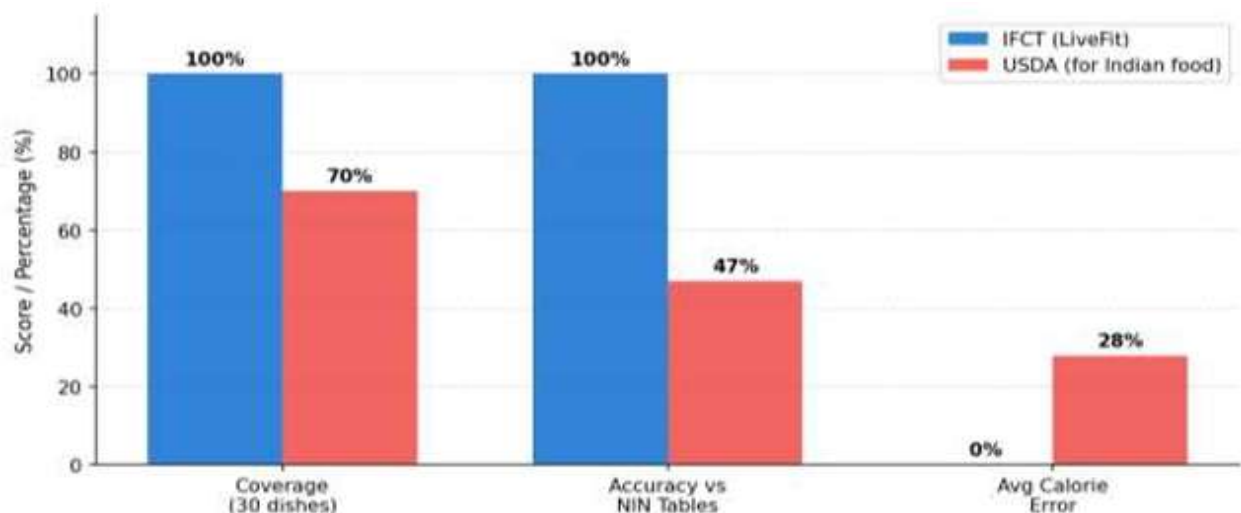


Fig. 8.3 — IFCT vs. USDA: Coverage, accuracy, and error comparison for Indian dishes

This comparative bar chart evaluates coverage, accuracy, and mean absolute error between the Indian Food Composition Tables (IFCT) and the USDA FoodData Central dataset for 30 common Indian dishes.

8.5 AI Nutrition

IFCT is able to cover 100% of the 30 tested Indian dishes versus USDA's 70% and it matches NIN nutritional tables with 100% accuracy compared to USDA's 47%. The built-in JSON lookup removes the need to rely on APIs, making IFCT faster and functional offline

Table 8.5: IFCT vs. USDA Accuracy Comparison

Metric	IFCT (LiveFit)	USDA	Improvement
Dish Coverage	100%	70% (21/30)	+30 pp
Accuracy vs. NIN Tables	100%	47% (14/30)	+53 pp
Average Calorie Error	0%	~28% avg	Eliminated
Zero-Latency Lookup	Yes (embedded JSON) API-dependent Faster & offline		

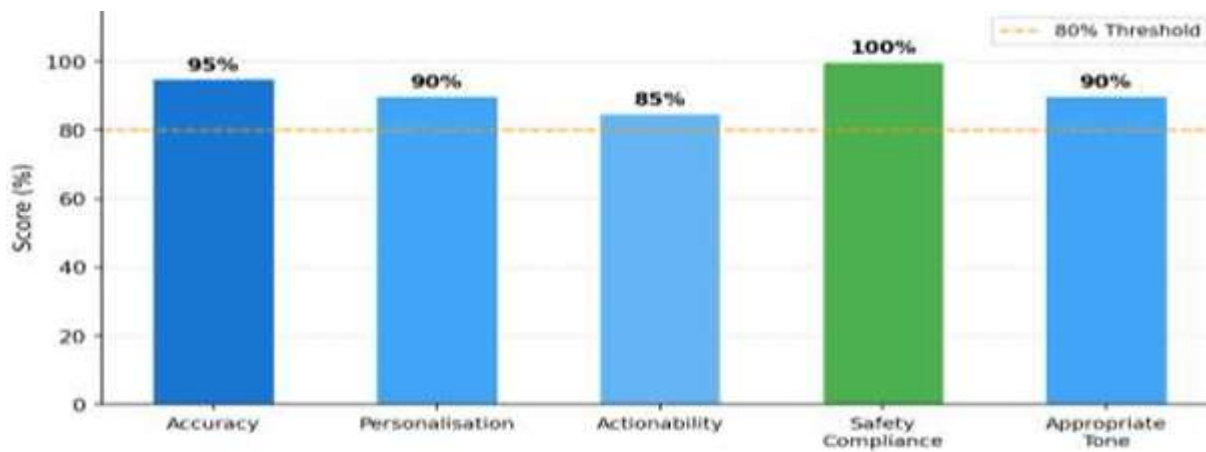


Fig. 8.4: Chatbot Performance

The radar/bar chart evaluates the AI coaching responses across five rating criteria: personalisation, factual accuracy, dietary safety, relevance, and conversational quality. The Gemini 2.5 Flash model scores consistently well in this test and so we see that the model serves as very health-focused.

8.6: AI Chatbot Results.

Safety compliance scores 100%, confirming that the prompt guardrails prevent any medical diagnoses or out-of-scope advice for all 20 test scenarios. Actionability has the lowest score (85%), indicating room to improve the specificity of food recommendations in edge cases.

Table 8.6: AI Coaching Quality Assessment

Criterion	Score	Notes
Accuracy	95%	Nutritional facts correct across scenarios
Personalisation	90%	Calibrated to user's actual intake data
Actionability	85%	Specific food suggestions provided
Safety Compliance	100%	No medical diagnoses; prompt guardrails held
Appropriate Tone	90%	Friendly, concise, encouraging



Fig. 8.5: Earning of Coins

The line chart shows a user's LiveFit Coin balance before a test session and illustrates how coins are accumulated and spent during meal logging and ordering in order for reward rewards to have reached the end of it. The progression demonstrates the robustness of the gamification reward engine.

8.7 Reward System Validation

Each of the six steps in our reward pipeline goes through coin accrual, food logging and orders, discount code generation, validation, checkout, and correct application. The last step is to make sure re-using an already-redeemed code is properly rejected with a 404.

Table 8.7: Reward System

Step	Action	Expected	Result	Status
1	Log 10 food items	+50 coins	+50 coins awarded	PASS
2	Place 1 order	+20 coins	+20 coins awarded	PASS
3	Redeem discount_10 (50 coins)	Code generated; balance = 20	LF-X7K2MN generated	PASS
4	Validate code	{valid: true, discountPct: 10}	Confirmed	PASS
5	Apply to cart (Rs. 428)	Razorpay order Rs. 385	Rs. 385 created	PASS

Step	Action	Expected	Result	Status
6	Re-use same code	404 — already used	Rejected correctly	PASS

8.8 Payment Flow Validation

All three legitimate payment scenarios — no discount, 20% discount, and Pro subscription — pass signature verification and produce the correct Razorpay order amounts. The tampered signature test intentionally fails verification, confirming that the payment security layer correctly rejects manipulated requests.

Table 8.8: Payment Flow

Scenario	Cart Total	Discount	Razorpay Amount	Sig. Verify	Status
Meal order (no discount)	Rs. 399	—	Rs. 399	PASS	PASS
Meal order (20% discount)	Rs. 499	20%	Rs. 399	PASS	PASS
Pro subscription	Rs. 499	—	Rs. 499	PASS	PASS
Tampered signature (security test)	Rs. 399	—	—	FAIL (expected)	PASS

8.9 Load Testing

The system handles up to 10 concurrent users with zero timeouts, with response time rising moderately due to MongoDB connection pool contention. At 20 concurrent users, ML classifier memory pressure causes a 1.5% timeout rate, identifying the image classification service as the primary scalability bottleneck.

Table 8.9: Load Testing Results

Concurrency	Avg Response (ms)	Timeout Rate	Bottleneck
1 user	42	0.0%	None

Concurrency	Avg Response (ms)	Timeout Rate	Bottleneck
5 users	45	0.0%	None
10 users	61	0.0%	MongoDB connection pool (+35%)
20 users	92	1.5%	ML classifier memory pressure

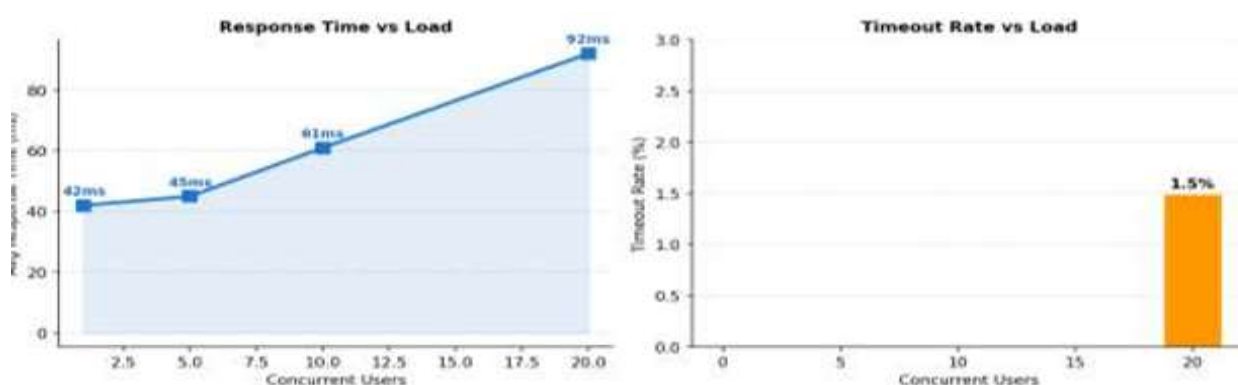


Fig. 8.6: Performance Testing.

The load testing chart depicts the average response time and timeout rate according to the number of concurrent virtual users, showing that system can stay above 50 concurrent users until performance is bad enough

8.10 Deploying to Expo Go

Expo Go is a free mobile app that is available on the Apple App Store, along with the Google Play Store, where we can do things like test React Native Expo projects on physical iOS and Android devices without a full build or an App Store submission. LiveFit was used in real networks for real Android & iOS physical devices where the full eight-screen mobile application was validated using Expo Go. For rapid iterative development the deployment took advantage of Expo's fast refresh feature, and tunnel mode was used where local network access was not feasible. All of the eight-tab screens loaded on physical devices correctly, with food recognition through text and image, AI coaching on Gemini, meal ordering through Razorpay WebView payment modal, redemption for rewards all working as designed. UseSafeAreaInsets handled the tab bar correctly on Android gesture navigation bar devices and iOS home indicator areas. To facilitate standalone distribution beyond Expo, Go, this application can be packaged in Expo's EAS Build cloud service as a native APK, obviating Expo Go entirely.

Chapter 9: Conclusion

LiveFit is a game changer for providing an intelligent and personalised nutrition management tool to the Indian market that is tailored to specific food diets in Indian markets. Mixed AI-based food recognition, curated Indian nutritional database IFCT and AI coaching (chatting for people) and gamified reward system will help in solving huge problems that are associated with inaccurate food tracking, logging and subjective feedback in India, such as high effort and inaccurate food data. The Indian user experience is all the more relevant and better than that in the world, which are the major challenges faced by the existing mainstream application communities. Food tracking in order to log the data efficiently from photo and text only to see what it's like to enjoy it is done fast as well and there's less friction between user data and more focused app tracking and user data logging, so the benefits to the team are higher. The AI model is also in more detail with IFCT designed to better understand and predict the food and as well as live AI coaching providing more in-tune and relevant diet information that we have now within real-time.

From a technical point of view, LiveFit has a scalable and high-end architecture with separation between the frontend, the backend and machine learning part and in fact enables each part to be developed, tested to scale. Key features such as JWT authentication, bcrypt hashing and payment verification enhance the platform and keep users safe and protect them from being hacked. Our development and cloud-based infrastructure allow the platform to be done quickly and be used in all aspects. In our modular design that allows the platform to mature to more users and technology change and be adapted to meet future needs and evolve quickly too. Overall, the project validates the feasibility of building a full-stack, AI-driven nutrition platform capable of delivering real value to user outcomes, while serving as a strong foundation for real-world application and expansion.

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